Pac-Man Game

1.1 Map Function

$$\text{map}(x,y) : M \subset \mathbb{N}^2 \;\rightarrow\; \{0,1,2,3,4\},$$
$$M \;=\; \Big\{(x,y) \in \mathbb{N}^2,\; x \in \{0,1,\ldots,w-1\},\; y \in \{0,1,\ldots,h-1\},\; w,h \in \mathbb{N}\Big\}$$
$$w = \text{map width},\; h = \text{map height}$$

1.2 State

$$x_t \in X_t \;=\; \Big\{\big(x_p,\, y_p,\, x_{g_1},\, y_{g_1},\, \ldots,\, x_{g_n},\, y_{g_n},\, c_1,\, x_{c_1},\, y_{c_1},\, c_2,\, x_{c_2},\, y_{c_2},\, \ldots,\, c_m,\, x_{c_m},\, y_{c_m}\big) \;\Big|$$
$$c_i \in \{0,1\}, \quad (x_{c_i}, y_{c_i}) \in \text{map}^{-1}(\{2\}) \iff \text{map}(x_{c_i}, y_{c_i}) = 2,$$
$$(x_p, y_p) \in \text{map}^{-1}(\{0,2,3,4\}) \iff \text{map}(x_p, y_p) \neq 1,$$
$$(x_{g_i}, y_{g_i}) \in \text{map}^{-1}(\{0,2,3,4\}) \iff \text{map}(x_{g_i}, y_{g_i}) \neq 1\Big\}.$$
$$n = \text{number of ghosts},\; m = \text{number of candies}$$

1.3 Input

Admissible pac-man moves set:

$$U \;=\; \Big\{(0,0), (-1,0), (1,0), (0,-1), (0,1)\Big\}.$$

$$u_t \in U_t \;=\; \Big\{(d_x, d_y) \in U,\; \text{map}(x_p^t + d_x, y_p^t + d_y) \neq 1,\; x_p^t, y_p^t \in x_t\Big\}.$$

1.4 Uncertainty

Admissible ghosts moves set:

$$U_g \;=\; \Big\{(-1,0), (1,0), (0,-1), (0,1)\Big\}.$$

$$w_i^t \in U_{gi}^t = \Big\{(d_{xi}, d_{yi}) \in U_g,\ \text{map}(x_{gi}^t + d_{xi}, y_{gi}^t + d_{yi}) \neq 1,\ x_{gi}^t, y_{gi}^t \in x_t\Big\} \forall\ i \in \{1, 2, \ldots, n\}$$

$$\text{manhattan}((x_1, y_1), (x_2, y_2)) = ||(x_1 - x_2)|| + ||(y_1 - y_2)||$$

$$w_1^t\ \ r.v.\ with\ \text{PMF}:$$
$$\text{P}\Big(w_1^t = d_i\Big) = \begin{cases} \frac{\frac{1}{(\text{manhattan}(d_i + r_{g1}, r_p))^{\text{power}} + 1}}{\sum_{j=1}^{c} \frac{1}{(\text{manhattan}(d_j + r_{g1}, r_p))^{\text{power}} + 1}} & \text{if } d_i \in U_{g1}^t \\ 0 & \text{otherwise} \end{cases}$$
$$d_j \in U_{g1}^t,\ r_{g1} = (x_{g1}^t, y_{g1}^t) \in x_t,\ r_p = (x_p^t, y_p^t) \in x_t,\ c = |U_{g1}^t|,\ \text{power} \in \mathbb{N}^+$$

$$w_i^t\ \forall\ i \in \{2, \ldots, n\}\ r.v.\ with\ \text{PMF}:$$
$$\text{P}\Big(w_i^t = d_i\Big) = \begin{cases} \frac{1}{c} & \text{if } d_i \in U_{gi}^t \\ 0 & \text{otherwise} \end{cases}$$
$$c = |U_{g1}^t|$$

1.5 Transition function

$$x_{t+1} = f(x_t, u_t, w_1^t, w_2^t, \ldots, w_n^t) = (x_p^t + d_x, y_p^t + d_y, x_{g1}^t + d_{x1}, y_{g1}^t + d_{y1}, \ldots, x_{gn}^t + d_{xn}, y_{gn}^t + d_{yn}, c^{t+1})$$
$$(d_x, d_y) = u_t,\ (d_{xi}, d_{yi}) = w_i^t\ \forall i \in \{1, \ldots, n\}$$
$$c^{t+1} = (c_1^{t+1}, x_{c1}, y_{c1}, \ldots, c_m^{t+1}, x_{cm}, y_{cm}),$$
$$c_i^{t+1} = \begin{cases} 0 & \text{if } x_{ci} = x_p^{t+1}\ \wedge\ y_{ci} = y_p^{t+1} \\ c_i^t & \text{otherwise} \end{cases}$$

1.6 Cost function

$$\text{eaten}(x_t, u_t) \begin{cases} \text{eat\_cost} & \text{if } \exists\ i \mid c_i^t = 1\ \wedge\ x_{ci} = x_p^t + d_x\ \wedge\ y_{ci} = y_p^t + d_y \\ 0 & \text{otherwise} \end{cases}$$

$$\text{lose}(x_t, u_t, w_1^t, \ldots, w_n^t) = \begin{cases} \text{lose\_cost} & \text{if } \exists\ i \mid x_p^t + d_x = x_{gi}^t + d_{xi}\ \wedge\ y_p^t + d_y = y_{gi}^t + d_{yi} \\ 0 & \text{otherwise} \end{cases}$$

$$\text{win}(x_t, u_t) = \begin{cases} \text{win\_cost} & \text{if } \sum_{i=1}^{m} c_i^t = 1\ \wedge\ c_i^t = 1\ \wedge\ x_{ci} = x_p^t + d_x\ \wedge\ y_{ci} = y_p^t + d_y \\ 0 & \text{otherwise} \end{cases}$$

Neighborhood Displacements:

$$N_g \ = \ \Big\{(-1,-1),(-1,0),(-1,1),(0,1),(1,1),(1,0),(1,-1),(0,-1)\Big\}.$$

$$\text{neighborhood}(x_t,u_t,w_i^t) =$$

$$= \begin{cases} \text{lose\_cost}/5 & \text{if } \exists \ n_g = (n_x,n_y) \in N_g |\ x_p^t + d_x = x_{gi}^t + d_{xi} + n_x \ \wedge \ y_p^t + d_y = y_{gi}^t + d_{yi} + n_y \\ 0 & \text{otherwise} \end{cases}$$

$$g_t(x_t,u_t,w_1^t,\ldots,w_n^t) = \text{move\_cost} + \text{eaten}(x_t,u_t) + \text{lose}(x_t,u_t,w_1^t,\ldots,w_n^t) + \text{win}(x_t,u_t)$$
$$+ \sum_{i=1}^{n} \text{neighborhood}(x_t,u_t,w_i^t)$$

1.7 States Enumeration

Total States Count:

$$\text{total\_states\_count} = \sum_{c=1}^{m} (free\_pos - c)\ free\_pos^n \binom{m}{c} + (m \cdot free\_pos^n)$$

Total Terminal States Count:

$$\text{total\_win\_states\_count} = m\ (free\_pos - 1)^n$$

$$\text{total\_lose\_states\_count} = \sum_{c=1}^{m}\sum_{g=1}^{n} (free\_pos - c)\ (free\_pos - 1)^{(n-g)} \binom{n}{g} \binom{m}{c} +$$
$$+ \sum_{g=1}^{n} m\ (free\_pos - 1)^{(n-g)} \binom{n}{g}$$

$$\text{total\_terminal\_states\_count} = \text{total\_win\_states\_count} + \text{total\_lose\_states\_count}$$